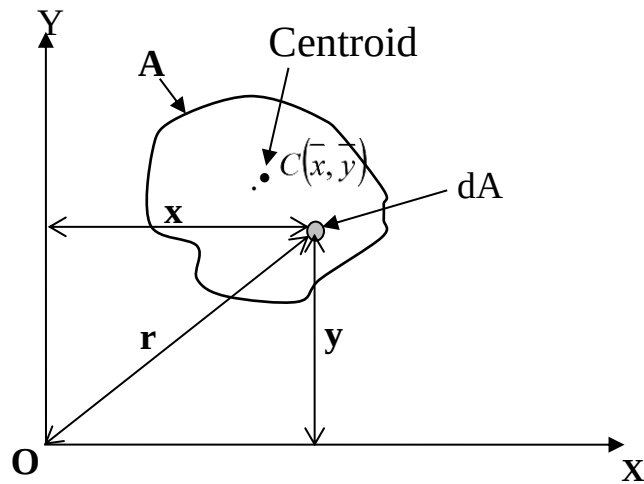


PROPERTIES OF PLANE AREAS

CENTROID OF AN AREA



Consider a plane area of any shape having an area 'A'. Take any set of orthogonal axes **OX**, **OY** which lie in the same plane. The position of a small elemental area 'dA' is given by its coordinates (x,y). Let 'r' be the distance of this elemental area from the origin 'O'. Therefore $r = \sqrt{(x^2 + y^2)}$.

DEFINITIONS:

$$\text{The first moment of area about x-axis} = \int_A y dA$$

$$\text{The first moment of area about y-axis} = \int_A x dA$$

The centroid, of an area refers to the point that defines the geometric centre for the area.

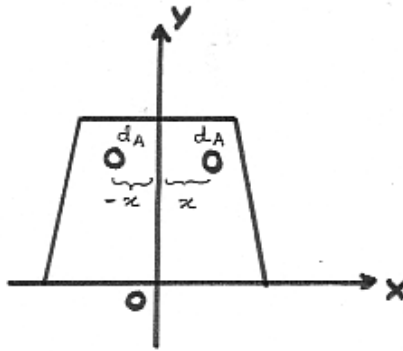
The coordinates of centroid $C \equiv (\bar{x}, \bar{y})$ can be determined using the following formulas.

$$A\bar{x} = \int_A x dA \qquad A\bar{y} = \int_A y dA$$

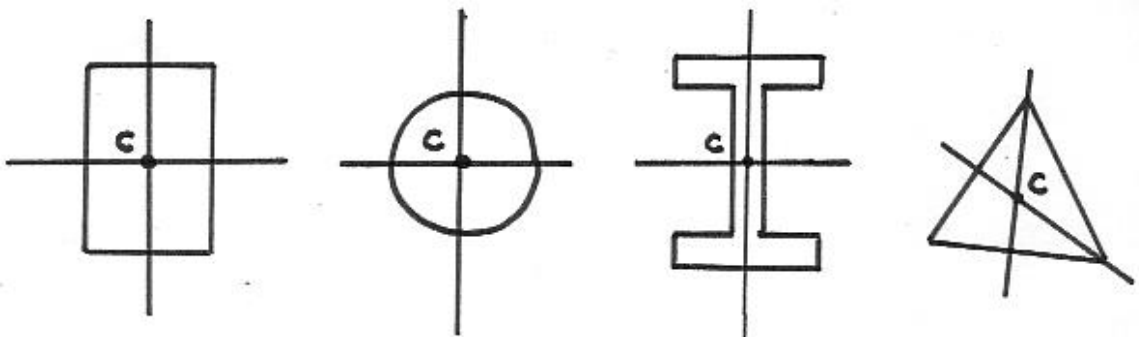
$$\bar{x} = \frac{\int_A x dA}{A} \qquad \text{and} \qquad \bar{y} = \frac{\int_A y dA}{A}$$

If the area under consideration possesses an axis of symmetry, then obviously the first moment of area about that axis is zero. (see figure below).

Therefore; $\int x.dA = 0$; $\therefore \bar{x} = 0$



This means that, if there exist an axis of symmetry for a particular area, then the centroid of that area is located on that axis of symmetry. It is also deduced from the above discussion, that is an area possesses two axes of symmetry, then the centroid of that area should be located at the intersection of the two axes of symmetry.



The second moment of area about x-axis; $I_{xx} = \int_A y^2 .dA$

The second moment of area about y-axis; $I_{yy} = \int_A x^2 dA$

The product of moment of area about x and y axes; $I_{xy} = \int_A x.y.dA$

The polar moment of the plane area about z-axis, which is perpendicular to the plane area;

I_{zz}

$$I_{zz} = J_0 = \int_A r^2 dA$$

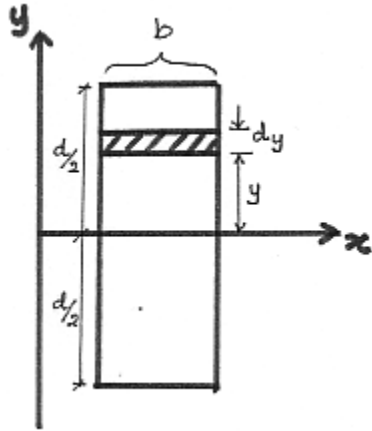
Since $r^2 = x^2 + y^2$; $I_{zz} = \int_A (x^2 + y^2) dA = \int_A x^2 dA + \int_A y^2 dA$

Therefore $I_{zz} = I_{xx} + I_{yy}$

Note:- I_{xx} , I_{yy} , I_{xy} and I_{zz} all have dimensions of $[L]^4$.

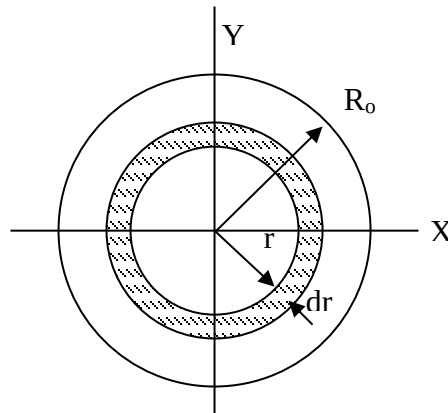
Examples:- Determine the second moment of areas of following sections.

Ex.1. Rectangular section, w.r.t. its centroidal axis;



$$\begin{aligned}
 I_{xx} &= \int y^2 dA \\
 I_{xx} &= \int_{-d/2}^{+d/2} (b \cdot dy) y^2 \\
 &= \frac{b}{3} \left[y^3 \right]_{-d/2}^{+d/2} \\
 &= \frac{b}{3} \frac{d^3}{4} = \frac{bd^3}{12}
 \end{aligned}$$

Ex.2. Polar moment of area for a circular section, w.r.t. Z-axis;



$$\begin{aligned}
 J_O = I_{zz} &= \int r^2 dA \\
 &= \int_0^{R_0} r^2 \cdot 2\pi \cdot dr \\
 &= 2\pi \left[\frac{r^4}{4} \right]_0^{R_0} = \frac{\pi}{2} R_0^4
 \end{aligned}$$

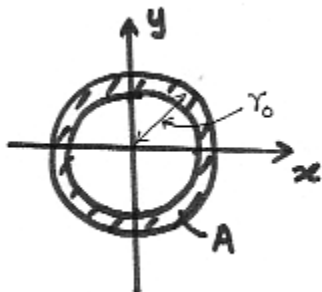
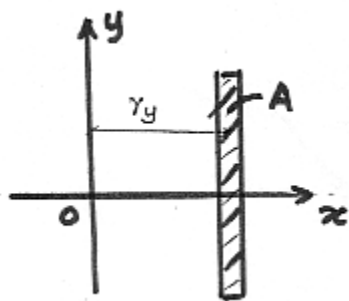
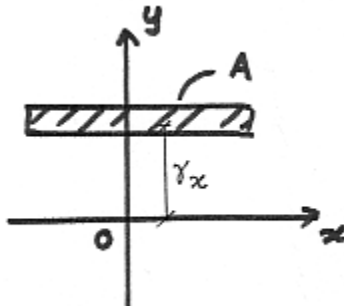
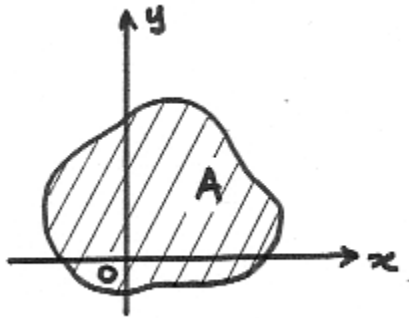
Ex.3. Second moment of area of the same section w.r.t. a diameter;

because of symmetry $I_{xx} = I_{yy}$

$$I_{zz} = I_{xx} + I_{yy} = 2I_{xx}$$

Therefore
$$I_{xx} = \frac{1}{2} I_{zz} = \frac{\pi}{4} R_0^4$$

Radius of Gyration of an Area



Let I_{xx} be the second moment of area of the plane area 'A' about x-axis

Then assume that the area 'A' is concentrated into a thin strip parallel to x-axis.

If this concentrated area 'A' is to have the same second moment of area I_{xx} about x-axis, the strip should be placed at a distance r_x from x-axis. Then this r_x is known as the

Radius of gyration about x-axis.

$$I_{xx} = A r_x^2$$

$$r_x = \sqrt{\frac{I_{xx}}{A}}$$

and similarly;

$$I_{yy} = A r_y^2$$

$$r_y = \sqrt{\frac{I_{yy}}{A}}$$

also

$$J_o = A r_o^2$$

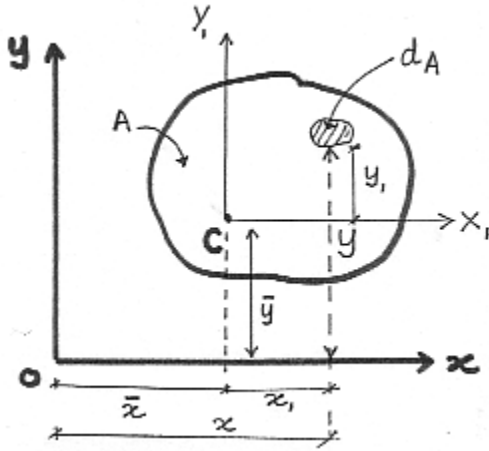
$$\text{i.e., } r_o = \sqrt{\frac{J_o}{A}}$$

The Parallel Axis Theorem

The centroid 'C' of a plane area has coordinates $(\bar{x}, \bar{y})$ with respect to x-axis and y-axis.

Let CX_1 and CY_1 be axes through the centroid, parallel to OX and OY axes respectively.

Then;



Proof

$$\begin{aligned} I_{xx} &= \int_A y^2 \cdot dA = \int_A (\bar{y} + y_1)^2 \cdot dA = \int_A \bar{y}^2 \cdot dA + \int_A 2\bar{y}y_1 \cdot dA + \int_A y_1^2 \cdot dA \\ &= \bar{y}^2 \int_A dA + 2\bar{y} \int_A y_1 \cdot dA + \int_A y_1^2 \cdot dA \end{aligned}$$

since for centroidal axis; $\int_A y_1 dA = 0$

$$I_{xx} = A\bar{y}^2 + 0 + I_{x_1x_1} = I_{x_1x_1} + A\bar{y}^2$$

Similarly I_{yy} can be obtained.

$$\begin{aligned} I_{xy} &= \int_A x \cdot y \cdot dA = \int_A (\bar{x} + x_1)(\bar{y} + y_1) dA = \int_A (\bar{x} \cdot \bar{y} + \bar{x} \cdot y_1 + x_1 \cdot \bar{y} + x_1 \cdot y_1) dA \\ &= \bar{x} \cdot \bar{y} \int_A dA + \bar{x} \int_A y_1 dA + \bar{y} \int_A x_1 dA + \int_A x_1 y_1 dA \end{aligned}$$

since for centroidal axes $\int_A y_1 dA = 0$ and $\int_A x_1 dA = 0$

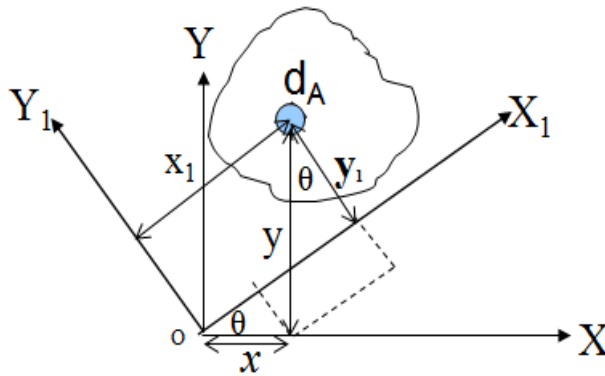
$$I_{xy} = \bar{x} \cdot \bar{y} \cdot A + I_{x_1y_1}$$

The Perpendicular Axis Theorem

This states that the polar moment of area of an area about a point is equal to the sum of the second moments of areas about two perpendicular axes through that point and lying in the plane.

$$I_{zz} = I_{xx} + I_{yy} \quad (\text{see page 3})$$

The Transformation Law (Second Moment of Area about Inclined Axes)



Given I_{xx} , I_{yy} , I_{xy} about axes OX, OY.

$$x_1 = x \cos \theta + y \sin \theta$$

$$y_1 = y \cos \theta - x \sin \theta$$

$$I_{x_1 x_1} = \int_A y_1^2 . dA = \int_A (y \cos \theta - x \sin \theta)^2 . dA$$

$$I_{x_1 x_1} = \cos^2 \theta \int y^2 . dA + \sin^2 \theta \int x^2 . dA - 2 \sin \theta \cos \theta \int xy . dA$$

$$= I_{xx} \cos^2 \theta + I_{yy} \sin^2 \theta - 2 \sin \theta \cos \theta . I_{xy}$$

$$= I_{xx} \left(\frac{1 + \cos 2\theta}{2} \right) + I_{yy} \left(\frac{1 - \cos 2\theta}{2} \right) - \sin 2\theta . I_{xy}$$

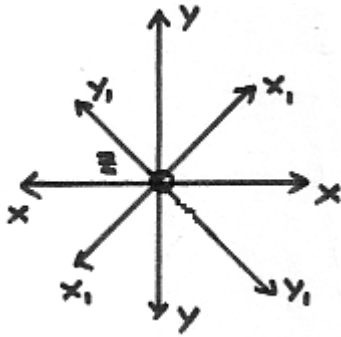
$$I_{x_1 x_1} = \left(\frac{I_{xx} + I_{yy}}{2} \right) + \left(\frac{I_{xx} - I_{yy}}{2} \right) \cos 2\theta - I_{xy} \sin 2\theta \dots\dots\dots [1]$$

Similarly, it can be proved that;

$$I_{y_1 y_1} = \left(\frac{I_{xx} + I_{yy}}{2} \right) - \left(\frac{I_{xx} - I_{yy}}{2} \right) \cos 2\theta + I_{xy} \sin 2\theta \dots\dots\dots [2]$$

and
$$I_{x_1 y_1} = \left(\frac{I_{xx} - I_{yy}}{2} \right) \sin 2\theta + I_{xy} \cos 2\theta \dots\dots\dots [3]$$

By adding equations [1] and [2];



$$I_{x1x1} + I_{y1y1} = I_{xx} + I_{yy} = I_{zz}$$

Principal Axes

The perpendicular axes through a selected point, about which the product of moment of area is zero are known as principal axes; i.e.

$$I_{x1y1} = 0$$

Directions of principal axes are obtained by equating the equation [3] above to zero;

$$I_{x1y1} = \left(\frac{I_{xx} - I_{yy}}{2} \right) \sin 2\theta + I_{xy} \cos 2\theta = 0$$

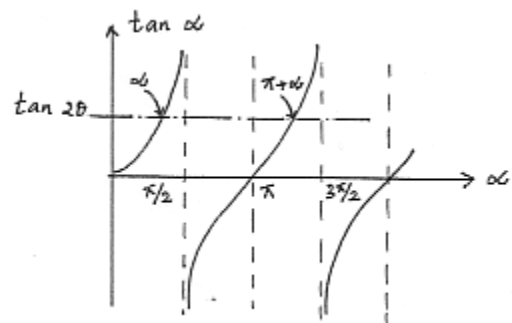
$$\tan 2\theta = -\frac{2I_{xy}}{(I_{xx} - I_{yy})}$$

we can find two solutions for this between 0 and 2π ;

$$2\theta = \alpha \quad \text{or} \quad 2\theta = \alpha + \pi$$

i.e

$$\theta = \frac{\alpha}{2} \quad \text{or} \quad \theta = \frac{\alpha + \pi}{2} = \frac{\alpha}{2} + \frac{\pi}{2}$$

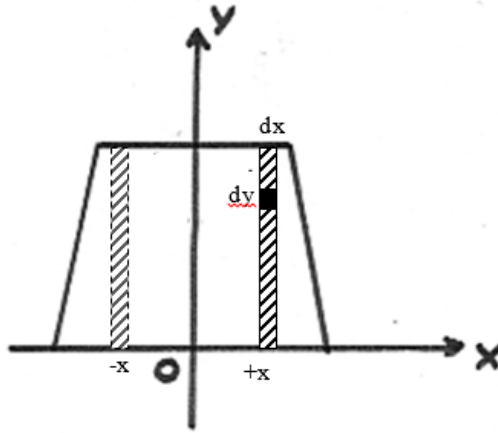


This means that the two principal axes are perpendicular to each other. The second moments of area along these two axes are known as **principal second moments of area**.

A principal axis through the centroid is known as a **centroidal principal axis**.

From the previous discussions, as $I_{xy} = \int x.y.dA = 0$, w.r.t. any axis of symmetry; any axis of symmetry is a principal axis. However, the reverse is not always true.

Note: $I_{xy} = \int x.y.dA = \int x.\{\int y.dy\}dx = 0$ for an axis of symmetry (e.g. y axis)



Principal Second Moment of Area

$I_{x1x1} + I_{y1y1} = I_{xx} + I_{yy} = I_{zz}$ is a constant (it is independent of θ)

Therefore when I_{x1x1} is a maximum, I_{y1y1} will be a minimum and vice versa.

To find the maximum I_{x1x1} ; $\frac{d}{d\theta}(I_{x1x1}) = 0$

from the equation [1] above;

$$\frac{d}{d\theta}(I_{x1x1}) = -2\left(\frac{I_{xx} - I_{yy}}{2}\right)\sin 2\theta - 2I_{xy}\cos 2\theta = 0$$

$$\tan 2\theta = -\frac{2I_{xy}}{(I_{xx} - I_{yy})}$$

This shows that the directions of the principal axes are also the directions along which the 2nd moments of areas are maximum and minimum. The second moments of area along these two axes are known as **principal second moments of area**. Value of I_{max} is given by;
From equation [1],

$$I_{max} = \left(\frac{I_{xx} + I_{yy}}{2}\right) + \left(\frac{I_{xx} - I_{yy}}{2}\right)\cos 2\theta - I_{xy}\sin 2\theta$$

Where $\tan 2\theta = -\frac{2I_{xy}}{(I_{xx} - I_{yy})}$

$$I_{\max} - \left(\frac{I_{xx} + I_{yy}}{2}\right) \sec 2\theta = \left(\frac{I_{xx} - I_{yy}}{2}\right) - I_{xy} \tan 2\theta$$

$$= \left(\frac{I_{xx} - I_{yy}}{2}\right) - I_{xy} \frac{(-2I_{xy})}{(I_{xx} - I_{yy})}$$

$$I_{\max} - \left(\frac{I_{xx} + I_{yy}}{2}\right) \sec 2\theta = \frac{(I_{xx} - I_{yy})^2 + 4I_{xy}^2}{2(I_{xx} - I_{yy})}$$

By substituting for $\sec 2\theta = \sqrt{\tan^2 2\theta + 1}$; where $\tan^2 2\theta = \frac{(I_{xx} - I_{yy})^2 + 4I_{xy}^2}{(I_{xx} - I_{yy})^2}$

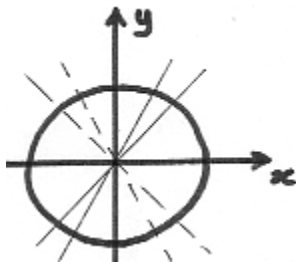
We have, $I_{\max} - \left(\frac{I_{xx} + I_{yy}}{2}\right) = \frac{1}{2} \sqrt{(I_{xx} - I_{yy})^2 + 4I_{xy}^2}$

$$I_{\max} = \left(\frac{I_{xx} + I_{yy}}{2}\right) + \sqrt{\left(\frac{I_{xx} - I_{yy}}{2}\right)^2 + I_{xy}^2}$$

Similarly $I_{\min} = \left(\frac{I_{xx} + I_{yy}}{2}\right) - \sqrt{\left(\frac{I_{xx} - I_{yy}}{2}\right)^2 + I_{xy}^2}$

Multiple Principal Axes

Special case of a plane area.



$$I_{xy} = 0;$$

$$I_{xx} = I_{yy} = I_O;$$

Directions of principal axes; $\tan 2\theta = \frac{0}{0} ???$

But for all values of θ ;

$$I_{x_1x_1} = I_{y_1y_1} = I_O \quad \text{and} \quad I_{x_1y_1} = 0$$

Therefore, whatever the direction is, every axis is a principal axis and the 2nd moment of area is the same for each of those axes. Therefore, if a plane area has more than 2 principal axes (axes of symmetry) through a point, then every axis through that point must be a principal axis and the 2nd moment of area is the same for every axis.

Examples

Consider the rectangular section in Figure Q.5. What is the second moment of area about the **diagonal axis AC** shown? You can use the following formulae (with symbols having usual meanings):

$$I_{x_1x_1} = \left(\frac{I_{xx} + I_{yy}}{2} \right) + \left(\frac{I_{xx} - I_{yy}}{2} \right) \cos 2\theta - I_{xy} \sin 2\theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

(a) 11.52 cm⁴ (b) 12.5 cm⁴ (c) 13.48 cm⁴ (d) 9 cm² (e) 16 cm⁴

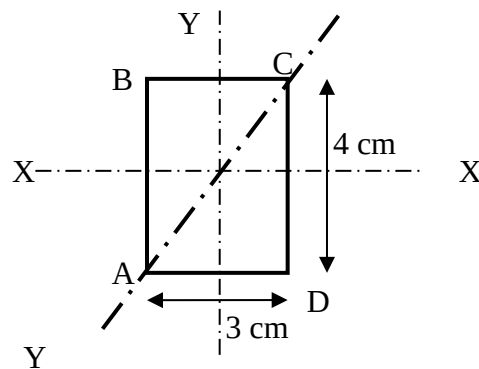
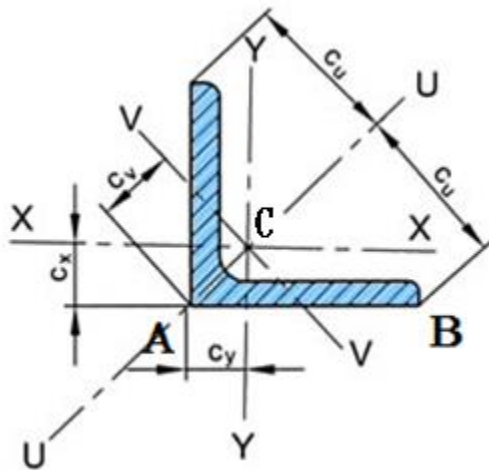


Figure Q.5

- Q.3. The section properties for a 75 mm x 75 mm x 8 mm equal angle section are given in Figure Q.(3-5) below. Axes X-X, Y-Y, U-U and V-V pass through the centroid, C.



$$\begin{aligned} c_u &= 5.30 \text{ cm} \\ c_v &= 3.03 \text{ cm} \\ c_x &= c_y = 2.14 \text{ cm} \\ I_{uu} &= 93.6 \text{ cm}^4 \\ I_{vv} &= 24.4 \text{ cm}^4 \\ A &= 11.4 \text{ cm}^2 \end{aligned}$$

Figure Q. (3-5)

What is the second moment of area of this section (**in cm⁴**) about an axis parallel to the U-U axis and passing through the point B?

- (A) 414** (B) 320 (C) 345 (D) 198 (E) 129

- Q.4. What is the second moment of area of the section (described in Q.3) in Figure Q.(3-5) above (**in cm⁴**) about axes X-X and Y-Y?

- (A) 69.2 (B) 118.0 **(C) 59.0** (D) 34.6 (E) 236.0

- Q.5. What is the polar moment of area of the section (described in Q.3) in Figure Q.(3-5) above (**in cm⁴**) about the point A?

- (A) Cannot be calculated (B) 118 **(C) 223** (D) 164 (E) 153